

Top 50 Statistics Interview Questions

For Product Analyst & Data Analyst Interviews

Prepared for Irfan Ansari | Quick-reference Q&A across the seven areas interviewers draw from most

How This List Is Organized

These 50 questions cover the statistics ground that comes up repeatedly across Product Analyst and Data Analyst interviews — not tool-specific (SQL/Python) questions, just the underlying statistical reasoning. They're grouped into seven areas:

- **A. Descriptive Statistics (Q1-7)** — summarizing data correctly
- **B. Probability Fundamentals (Q8-15)** — the building blocks under every statistical test
- **C. Hypothesis Testing (Q16-27)** — p-values, errors, and choosing the right test
- **D. A/B Testing & Experimentation (Q28-37)** — the practical, product-analyst-specific core
- **E. Regression & Relationships (Q38-44)** — correlation, regression, overfitting
- **F. Sampling, Estimation & Bias (Q45-48)** — where studies go wrong
- **G. Applied Reasoning (Q49-50)** — the "does this actually follow" questions

Answers are kept to the core idea in 2-4 lines — enough to answer confidently and then expand out loud with your own example if asked to go deeper.

A. Descriptive Statistics

Q1. When would you use mean vs. median vs. mode?

Mean for symmetric, non-skewed data without extreme outliers. Median when data is skewed or has outliers (e.g., income, revenue per user) since it isn't pulled by extremes. Mode for categorical data or to find the most common value (e.g., most-used feature).

Q2. What's the difference between variance and standard deviation?

Variance is the average squared deviation from the mean; standard deviation is its square root, which puts spread back in the original units — making it easier to interpret alongside the mean.

Q3. What is skewness, and how do you detect it?

Skewness measures asymmetry in a distribution. Right (positive) skew: long tail of high values (e.g., spend data with whales) — mean > median. Left (negative) skew: mean < median. Detect via histogram, skewness coefficient, or comparing mean to median.

Q4. What is kurtosis?

Kurtosis measures how heavy-tailed a distribution is relative to normal. High kurtosis ("leptokurtic") means more extreme outliers than a normal distribution would predict — relevant when deciding if normal-based tests are appropriate.

Q5. How do you detect and handle outliers?

Detect via IQR (1.5× beyond Q1/Q3), z-score (>3 SD from mean), or visually (box plot). Handling depends on cause: remove if it's a data error, cap/winsorize if it's real but extreme, or keep and use a robust metric (median) if outliers are meaningful (e.g., whale spenders).

Q6. Why do we divide by (n−1), not n, for sample variance?

Using the sample mean (instead of the true population mean) to compute deviations slightly understates variance. Dividing by (n−1) — Bessel's correction — corrects this bias so the sample variance is an unbiased estimator of the population variance.

Q7. What's the difference between a "high variance" metric and a "biased" metric?

High variance means estimates bounce around a lot from sample to sample but average out to the truth over many samples. Bias means the estimate is systematically off-target regardless of sample size. You can have low-variance-but-biased or high-variance-but-unbiased estimators — they're independent problems.

B. Probability Fundamentals

Q8. What is conditional probability?

The probability of event A given that event B has occurred: $P(A|B) = P(A \text{ and } B) / P(B)$. Example: probability a user churns given they had a failed payment, versus churn probability overall.

Q9. State Bayes' Theorem and give an example of why it matters.

$P(A|B) = P(B|A) \cdot P(A) / P(B)$. It matters because it lets you update a prior belief with new evidence correctly — e.g., a 99%-accurate fraud model still produces mostly false positives if actual fraud is rare (low prior), which naive intuition misses.

Q10. What's the difference between independent and mutually exclusive events?

Independent: one event's occurrence doesn't change the probability of the other ($P(A|B) = P(A)$). Mutually exclusive: the two events can't happen together ($P(A \text{ and } B) = 0$). Mutually exclusive events are actually strongly dependent — knowing one occurred tells you the other didn't.

Q11. What's the difference between a discrete and continuous probability distribution?

Discrete: takes specific countable values (e.g., number of purchases — binomial, Poisson). Continuous: takes any value in a range (e.g., session duration — normal, exponential), described by a probability density function rather than exact point probabilities.

Q12. When would you use a binomial distribution?

For the number of successes in a fixed number of independent yes/no trials with constant success probability — e.g., number of converters out of 1,000 visitors, each independently converting with the same probability p.

Q13. What are the key properties of a normal distribution?

Symmetric, bell-shaped, defined by mean and standard deviation; mean = median = mode. The 68-95-99.7 rule: ~68% of values fall within 1 SD of the mean, ~95% within 2 SD, ~99.7% within 3 SD.

Q14. When would you use a Poisson distribution?

For counting the number of events in a fixed interval of time/space, when events happen independently at a constant average rate — e.g., number of support tickets per hour, number of crashes per day.

Q15. What is the Central Limit Theorem and why does it matter for analytics?

The sampling distribution of the mean approaches a normal distribution as sample size grows, regardless of the underlying data's distribution. This is why t-tests/z-tests on averages are valid at scale even when raw per-user data (e.g., spend) is heavily skewed.

C. Hypothesis Testing

Q16. What's the difference between the null and alternative hypothesis?

Null (H_0): assumes no effect/no difference (the default, skeptical position). Alternative (H_1): the effect/difference you're trying to find evidence for. A test either rejects H_0 in favor of H_1 , or fails to reject H_0 — it never "proves" H_0 true.

Q17. Explain a p-value in plain language.

The probability of observing data at least as extreme as what you saw, assuming the null hypothesis is true. A small p-value (e.g., <0.05) means the observed result would be unlikely if there were really no effect.

Q18. What's a common misinterpretation of the p-value?

Thinking $p = 0.03$ means "3% chance the null hypothesis is true" or "97% chance the result is real." A p-value says nothing about the probability of the hypothesis itself — it's a statement about the data assuming the null is true.

Q19. What are Type I and Type II errors?

Type I (α): false positive — rejecting a true null (concluding an effect exists when it doesn't). Type II (β): false negative — failing to reject a false null (missing a real effect). They trade off against each other for a fixed sample size.

Q20. What is statistical power, and what increases it?

Power = $1 - \beta$, the probability of correctly detecting a real effect. Increased by larger sample size, larger true effect size, lower variance in the data, and a higher significance threshold (α) — though that raises Type I error risk.

Q21. How do you choose a significance level (α)?

0.05 is conventional, but the right choice depends on the cost of a false positive vs. false negative for the specific decision — a riskier or harder-to-reverse decision may warrant a stricter threshold (e.g., 0.01).

Q22. One-tailed vs. two-tailed test — when do you use each?

Two-tailed tests for "is there any difference" (effect could go either direction) — the standard, more conservative default. One-tailed only when you have a strong prior reason to only care about one direction (e.g., "did this feature increase, and only increase, retention") and commit to that before seeing the data.

Q23. T-test vs. Z-test — when do you use each?

Z-test when population variance is known or sample size is large ($n > 30$, CLT applies). T-test when population variance is unknown and estimated from the sample, especially with smaller samples — the t-distribution has heavier tails to account for that extra uncertainty.

Q24. When would you use a chi-square test?

To test association between two categorical variables (e.g., platform vs. churned/not-churned) via a contingency table, or to compare an observed categorical distribution against an expected one (goodness-of-fit).

Q25. What does ANOVA test, and why not just run multiple t-tests?

ANOVA tests whether the means of 3+ groups are significantly different in one combined test. Running pairwise t-tests instead inflates the overall Type I error rate (multiple comparisons problem) — ANOVA controls for that in a single test.

Q26. What is a confidence interval, and how do you interpret a 95% CI?

A range of plausible values for a population parameter. A 95% CI means: if you repeated the sampling process many times, 95% of the intervals constructed this way would contain the true value — it's not "95% probability the true value is in this specific interval."

Q27. What's the difference between statistical and practical significance?

Statistical significance means the effect is unlikely to be due to chance. Practical significance means the effect is large enough to matter for the business. With very large samples, tiny, meaningless effects can be statistically significant — always check effect size, not just the p-value.

D. A/B Testing & Experimentation

Q28. Walk me through designing an A/B test end-to-end.

Define a specific hypothesis → pick a primary metric + guardrails → choose the randomization unit → calculate sample size/duration from baseline rate, MDE, α , and power → run for full weekly cycles → check for sample ratio mismatch → analyze → decide and document.

Q29. How do you calculate the required sample size for a test?

Driven by baseline conversion rate, minimum detectable effect (MDE), significance level (α , usually 0.05), and desired power (usually 0.80): $n \propto (z_{\alpha/2} + z_{\beta})^2 \times \text{variance} / \text{MDE}^2$. Smaller MDE or higher variance both require a larger sample.

Q30. What is "peeking," and why is it a problem?

Checking results repeatedly and stopping as soon as you see significance. Each look is effectively another hypothesis test, so the true false-positive rate compounds well above the nominal 5%. Fix: pre-register duration/sample size, or use a sequential testing method designed for early stopping.

Q31. What are guardrail metrics, and why do you need them alongside a primary metric?

Metrics that must not regress even if the primary metric improves (e.g., crash rate, load time, other feature usage). They catch a feature that "wins" on its target metric while quietly damaging the broader experience.

Q32. What's the multiple comparisons problem, and how do you correct for it?

Testing many metrics/segments at $\alpha=0.05$ each raises the chance that at least one looks significant purely by chance. Corrections: Bonferroni (divide α by number of tests, conservative) or Benjamini-Hochberg/FDR (controls the expected proportion of false discoveries, less conservative).

Q33. What is the novelty effect in experimentation?

Users engage more with something simply because it's new, inflating early metrics regardless of the feature's real long-term value. Mitigate by running the test long enough to look past the initial novelty window and checking if the lift is sustained.

Q34. What is Sample Ratio Mismatch (SRM), and how do you detect it?

When the actual traffic split between control/treatment doesn't match the intended ratio (e.g., 48/52 instead of 50/50), often signaling a bugged assignment or logging issue that can invalidate the whole test. Detect with a chi-square goodness-of-fit test on the observed vs. expected group sizes.

Q35. How do you A/B test something with network effects or that can't be randomized per user?

Use switchback tests (alternate treatment over time periods) or geo/cluster-based randomization (randomize by region or server shard instead of individual user) to avoid contamination between treatment and control groups sharing the same environment.

Q36. What is CUPED, and why would you use it?

Controlled-experiment Using Pre-Experiment Data — a variance-reduction technique that adjusts the outcome metric using a covariate (typically the same metric's pre-experiment value) to reduce noise, letting you detect the same effect with a smaller sample or shorter test.

Q37. How could Simpson's Paradox show up in an A/B test result?

If the traffic mix shifts across time or segments during the test (e.g., a ramp-up rollout, or platform mix changing), the pooled result can reverse or mask what's actually happening in each subgroup. Always check the effect direction is consistent within key segments before trusting the aggregate.

E. Regression & Relationships

Q38. Correlation vs. causation — explain with an example.

Two variables can move together without one causing the other, due to a confounder. E.g., "players who reach level 10 spend more" — leveling doesn't cause spending; underlying engagement drives both. Causation requires a controlled experiment (or strong quasi-experimental design), not just observed correlation.

Q39. Pearson vs. Spearman correlation — what's the difference?

Pearson measures linear relationship strength between two continuous variables and is sensitive to outliers. Spearman measures monotonic relationship using ranked data, so it captures non-linear-but-consistent trends and is more robust to outliers.

Q40. What is R-squared, and what's a common misuse of it?

The proportion of variance in the outcome explained by the model. Common misuse: assuming a high R^2 means the model is "good" or causal — R^2 always increases (or stays flat) as you add more predictors, even irrelevant ones, so it doesn't penalize overfitting on its own (use adjusted R^2 instead).

Q41. What are the key assumptions of linear regression?

Linearity (relationship between X and Y is linear), independence of errors, homoscedasticity (constant error variance), normally distributed residuals, and no severe multicollinearity among predictors.

Q42. What is multicollinearity, and how do you detect/fix it?

When predictor variables are highly correlated with each other, making individual coefficient estimates unstable and hard to interpret. Detect via Variance Inflation Factor (VIF) or a correlation matrix; fix by removing/combining correlated predictors or using regularization (ridge/lasso).

Q43. When would you use logistic regression instead of linear regression?

When the outcome is binary/categorical (e.g., churned or not) rather than continuous. Logistic regression models the log-odds of the outcome as a linear function of predictors; coefficients are interpreted as changes in odds (exponentiate to get the odds ratio).

Q44. What is overfitting, and how do you prevent it?

A model that fits the training data's noise, not just its signal, and performs poorly on new data. Prevent with cross-validation to evaluate on held-out data, regularization (L1/L2), simplifying the model, or gathering more training data.

F. Sampling, Estimation & Bias

Q45. What's the difference between sampling bias and selection bias?

Sampling bias: the method of drawing the sample systematically favors certain outcomes (e.g., surveying only app power-users about satisfaction). Selection bias is the broader concept: any process (including who "opts in" or survives to be observed) that makes the analyzed group non-representative of the population you care about.

Q46. What's the difference between standard deviation and standard error?

Standard deviation measures spread of individual data points around the mean. Standard error measures the precision of the sample mean itself as an estimate of the population mean, and shrinks as sample size grows ($SE = SD / \sqrt{n}$).

Q47. What is regression to the mean?

Extreme observations (a record-high or record-low metric day) tend to be followed by more average ones purely due to natural variance, not because of any intervention. Trap: launching a fix right after an unusually bad data point and crediting the fix when the metric naturally reverts.

Q48. What is survivorship bias? Give an example.

Drawing conclusions only from the subset that "survived" a process while ignoring those that didn't, skewing the result. Classic example: analyzing satisfaction only among current users of a feature ignores everyone who tried it and left, overstating how well-liked it is.

G. Applied Reasoning

Q49. If Hospital A has a higher success rate than Hospital B in every category of surgery, is A's overall rate always higher? Simpson's Paradox

No. If the two hospitals treat very different mixes of easy vs. hard cases, the overall (weighted) rate can reverse even though A wins in every individual category — because the category weights, not just the category rates, drive the total. Always check subgroup mix before trusting an aggregate comparison.

Q50. Three doors, one prize. You pick one; the host reveals a non-prize door and offers a switch. Should you switch?Monty Hall

Yes — switching wins 2/3 of the time, staying wins only 1/3. Your original pick had a 1/3 chance when made; the host's guaranteed reveal of a non-prize door doesn't change that, it just concentrates the remaining 2/3 probability onto the door you didn't pick.